wrongly, that the X-ray sky would be dominated by objects already known in the optical sky—objects like the Moon, planets, and ordinary stars that are poor emitters of X-rays. However, Sco X-1 and other X-ray stars soon to be discovered were not a type of object anyone had ever seen before. They were neutron stars and black holes, capturing gas from normal-star companions and heating it to high temperatures in the manner soon to be proposed by Zel'dovich and Novikov (Figure 8.5 above). To deduce that this was indeed the nature of the observed X-ray stars, however, would require another decade of hand-in-hand hard work by experimenters like Friedman and Giacconi and theorists like Zel'dovich and Novikov.

Giacconi's 1962 detector was exceedingly simple (Figure 8.6a): an electrified chamber of gas, with a thin window in its top face. When an X-ray passed through the window into the chamber, it knocked electrons off some of the gas's atoms; and those electrons were pulled by an electric field onto a wire, where they created an electric current that announced the X-ray's arrival. (Such chambers are sometimes called *Geiger counters* and sometimes *proportional counters.)* The rocket carrying the chamber was spinning at two rotations per second and its nose slowly swung around from pointing up to pointing down. These motions caused the chamber's window to sweep out a wide swath of sky, pointing first in one direction and then another. When pointed toward the constellation Scorpius, the chamber recorded many X-ray counts. When pointed elsewhere, it recorded few. However, because X-rays could enter the chamber from a wide range of directions, the chamber's estimate of the location of Sco X-1 on the sky was highly uncertain. It could report only a best-guess location, and a surrounding 90-degree-wide *error box* indicating how far wrong the best guess was likely to be (see Figure 8.6b).

To discover that Sco X-1 and other X-ray stars soon to be found were in fact neutron stars and black holes in binary systems would require error boxes (uncertainties in position on the sky) a few minutes of arc in size or smaller. That was a very tall order: a 1000-fold improvement in angular accuracy.

The needed improvement, and much more, came step by step over the next sixteen years, with several teams (Friedman's, Giacconi's, and others) competing at each step of the way. A succession of rocket flights by one team after another with continually improving detectors was followed, in